THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 24 cm	Image forms at C = 2f.
2	**(c)** -1 D	P = 1/(-1 m).
3	**(d)** 2×10^{19}	n = 3.2 / 1.6×10^{-19}.
4	**(b)** is reversed	The field direction reverses.
5	**(c)** 0.5	0.75/1.5 = 0.5.
6	**(c)** a single red patch	Monochromatic light does not disperse.
7	**(d)** 2 ohm	R / n.
8	**(b)** to the south	By Fleming's left-hand rule: thumb points to the south.
9	**(d)** 40 cm	1/v = 1/20 - 1/40 = 1/40.
10	**(d)** dark	No scattering.
11	**(d)** 1100 W	P = V I.
12	**(c)** increases	Faster change gives a larger emf.
13	**(a)** 30 cm	Same size at C.
14	**(b)** +2 D	1/f = 1/25 - 1/50 = 1/50 gives +2 D.
15	**(c)** 2 A	I = 600 / 300.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	1/f = 1/v + 1/u = -1/30 - 1/30 (1); f = -15 cm (1).	1+1
17	1/f = -1/40 + 1/25 = 3/200 (1); f = 66.7 cm (1).	1+1
18	n = I / e = 0.8 / 1.6×10^{-19} (1); n = 5×10^{18} (1).	1+1
19	At the point of intersection there would be two directions of the field (1); The field has only one direction at a point (1).	1+1
20	v = c/n = 3×10^8 / 1.6 (1); v = 1.875×10^8 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	1/v = -1/15 + 1/20 = -1/60 (1); v = -60 cm; m = -v/u = -3 (1); Size 18 cm; real, inverted, magnified (1).	1+1+1
22	u = infinity, v = -1.5 m (1); f = -1.5 m (1); P = -0.67 D, concave lens (1).	1+1+1
23	R = 3 + 4 + 2 = 9 ohm (1); I = 9 / 9 = 1 A (1); V = I R = 4 V (1).	1+1+1
24	Galvanometer deflects when the magnet moves in (1); It deflects in the opposite direction when withdrawn (1); The magnetic field through the coil changes, inducing a current (1).	1+1+1
	OR: 1/v = -1/10 - 1/20 = -3/20 (1); v = -6.67 cm (1); m = v/u = 1/3; virtual, erect, diminished (1).	1+1+1
25	At noon light has a short path in the atmosphere (1); At sunset it travels a long path (1); Most blue is scattered away, so red remains (1).	1+1+1
	OR: R = rho L / A (1); Greater area gives lower resistance (1); Less heat I^2 R t, so the wire does not overheat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	1/v = -1/12 + 1/20 = -1/30 (1); v = -30 cm (1); m = -v/u = -1.5 (1); Image size = -4.5 cm (1); Real, inverted, magnified; rays: parallel and through F (1).	1+1+1+1+1
	OR: (a) Hypermetropia (1); (b) 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D (1); (c) Convex lens (1); Image forms at the near point (1).	1+1+1+1+1

27	(a) Parallel = 2 ohm; total = 6 ohm (1); (b) I = 12 / 6 = 2 A (1); (c) V = 2 x 2 = 4 V (1); (d) I in 6 ohm = 4/6 = 0.67 A (1); I in 3 ohm = 4/3 = 1.33 A (1).	1+1+1+1+1
	OR: Concentric circles around the wire (1); Doubling the current increases the field (1); Doubling the distance weakens the field (1); Right-hand thumb rule: thumb along current (1); Fingers curl in the direction of the field (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** larger than B	Nearer, stronger.
2	**(a)** 1.5×10^8 m/s	v = c/n.
3	**(b)** the refractive index for violet is higher	n is larger for shorter wavelengths.
4	**(d)** 8 ohm	400 / 50.
5	**(d)** perpendicular to the velocity	Force is always perpendicular to velocity.
6	**(d)** 3 cm (inverted)	v = 20 cm, m = -1.
7	**(b)** light travels a short path and little blue scatters	Colours combine.
8	**(d)** 484 ohm	V^2 / P.
9	**(b)** the magnet is at rest relative to the coil	No change in the field.
10	**(c)** 36 cm	1/v = -1/12 + 1/18 = -1/36.
11	**(c)** -40 cm	f = 1/P.
12	**(b)** 36 J	W = V Q.
13	**(c)** a north pole	Anticlockwise looks like north.
14	**(b)** $\sqrt{2}$	sin45/sin30 = (1/$\sqrt{2}$)/(1/2) = $\sqrt{2}$.
15	**(a)** red	n is smallest for red.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	His near point is beyond 25 cm (1); Only at the larger distance the image falls on the retina (1).	1+1
17	1/R = 1/20 + 1/5 = 5/20 (1); R = 4 ohm (1).	1+1
18	Force is zero (1); Force is maximum only when the wire is perpendicular to the field (1).	1+1
19	It always diverges rays (1); Refracted rays appear to meet only when extended backwards (1).	1+1
20	Less air above scatters less light (1); The sky looks deep blue/dark (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Motor converts electrical energy into mechanical energy; generator does the reverse (1); Motor works on the force on a current-carrying conductor in a field; generator on electromagnetic induction (1); Motor uses a split-ring commutator; an AC generator uses slip rings (1).	1+1+1
22	Both give real, inverted, same-sized images (1); Mirror image forms on the same side of the object (1); Lens image forms on the opposite side (1).	1+1+1
23	25 cm (1); At smaller distances the lens cannot become thick enough (1); The image would be blurred (1).	1+1+1
24	Required equivalent R = 220 / 5 = 44 ohm (1); n resistors in parallel give 220 / n (1); n = 5 (1).	1+1+1
	OR: Straight wire: concentric circles (1); Solenoid: field like a bar magnet (1); Solenoid's field is uniform inside (1).	1+1+1
25	1/v = 1/20 + 1/20 = 1/10 (1); v = +10 cm (1); m = -v/u = 0.5; virtual, erect, diminished (1).	1+1+1
	OR: 1/f = -1/75 + 1/25 = 2/75 (1); f = 37.5 cm (1); P = +2.67 D (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(a) Water, kerosene, glass, diamond (1); (b) v = 3×10^8/2.42 (1); v = 1.24×10^8 m/s (1); (c) Diamond (1); Greater n means greater bending towards the normal (1).	1+1+1+1+1
	OR: u = infinity, v = -80 cm (1); f = -80 cm (1); P = -1.25 D (1); Near point is 25 cm, normal (1); No lens needed for reading, but a concave lens for distance (1).	1+1+1+1+1

27	H = I^2 R t (1); H = 4 x 50 x 600 (1); H = 1.2×10^5 J (1); In calories: 1.2×10^5 / 4.2 (1); = 28571 cal (1).	1+1+1+1+1
	OR: Mains: live, neutral and earth wires (1); Appliances are connected in parallel (1); Geyser current = 2000 / 220 = 9.1 A (1); More than 5 A, so needs the 15 A line (1); Fuse in the live wire melts for excess current (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 1 ohm	1/2 + 1/3 + 1/6 = 1.
2	**(a)** zero	No force parallel to the field.
3	**(d)** 20 cm	1/u = 1/v - 1/f = 1/60 - 1/15 = -1/20.
4	**(a)** blue light	Short wavelength scatters more.
5	**(a)** 1 kWh	2 x 0.5.
6	**(b)** a larger induced emf	More turns, more emf.
7	**(b)** 12 cm behind the mirror	1/v = 1/20 + 1/30 = 1/12.
8	**(a)** in front of the retina	Because the eye is too strong.
9	**(d)** 10 V	V = I R.
10	**(b)** increases	Proportional to the current.
11	**(a)** water	Water has a lower n.
12	**(d)** white light	Recombination.
13	**(a)** 1 A	I = 16 / 16.
14	**(b)** Reversing the current	Direction changes.
15	**(a)** virtual, erect and three times larger	Positive m means virtual and erect.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Voltage gets divided among the appliances (1); If one fails the others also stop (1).	1+1
17	100 times per second (1); 6000 times per minute (1).	1+1
18	m = -v/u = -(7.5)/(-15) (1); m = +0.5 (1).	1+1
19	f = -2 m (1); P = -0.5 D (1).	1+1
20	Rs = 25 ohm (1); I = 5 / 25 = 0.2 A (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	1/R = 1/4 + 1/6 + 1/12 = 6/12 (1); R = 2 ohm (1); I = 6 / 2 = 3 A (1).	1+1+1
22	(a) Increases with current (1); (b) Increases with turns per unit length (1); (c) Increases with a soft iron core (1).	1+1+1
23	1/v = 1/8 - 1/12 = 1/24 (1); v = 24 cm, m = -2 (1); Height 8 cm; real, inverted (1).	1+1+1
24	The second prism reverses the dispersion (1); Colours combine to form white light (1); Shows the prism only separates existing colours (1).	1+1+1
	OR: R = V^2 / P (1); 60 W bulb: R = 806.7 ohm (1); 100 W bulb: 484 ohm, so 60 W has more resistance (1).	1+1+1
25	Emf is produced only when the coil cuts the field lines (1); A stationary coil has no change in flux (1); So the current stops when the rotation stops (1).	1+1+1
	OR: Focal length increases (1); Relative refractive index between lens and water decreases (1); So the lens bends light less (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	f = +20 cm (1); 1/v = 1/20 + 1/20 = 1/10, v = +10 cm (1); m = -v/u = +0.5 (1); Height 2.5 cm (1); Virtual, erect, diminished behind the mirror (1).	1+1+1+1+1
	OR: (a) Myopia (1); (b) u = infinity, v = -40 cm (1); f = -40 cm (1); P = -2.5 D (1); (c) Concave lens diverges rays so a distant object appears at 40 cm (1).	1+1+1+1+1

27	(a) R = 60 ohm (1); (b) I = 12 / 60 = 0.2 A (1); (c) P1 = I^2 R = 0.4 W (1); P2 = 0.8 W (1); P3 = 1.2 W (1).	1+1+1+1+1
	OR: Wind many turns of insulated copper wire over a soft iron rod (1); Connect to a battery (1); Strength increases with current and with number of turns (1); Soft iron core increases it (1); Loses magnetism when current stops, so temporary (1).	1+1+1+1+1

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